



Original Investigation | Urology

Racial Differences in Adverse Events After Androgen Deprivation in Veterans With Prostate Cancer

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Abstract

IMPORTANCE Androgen deprivation therapy (ADT) for prostate cancer (PC) has been linked to adverse events, such as cardiac events and osteoporosis. Whether these associations differ by race is unknown.

OBJECTIVE To assess the association between ADT and time to cardiovascular and bone-related adverse events across racial groups.

DESIGN, SETTING, AND PARTICIPANTS This retrospective, nationwide, population-based, cohort study used electronic health records from the Veterans Affairs Health Care System (VAHCS) database from January 2001 to December 2021, with follow-up until death or study end. Participants included veterans with PC. Patients undergoing bilateral orchiectomy, receiving gonadotropin-releasing hormone antagonists, or lacking data on race and ethnicity were excluded. Data analysis was performed from January to March 2023.

EXPOSURE Luteinizing hormone–releasing hormone agonists (leuprolide, goserelin, or triptorelin) use as a time-dependent variable and first-line systemic therapy.

MAIN OUTCOMES AND MEASURES The primary outcome was time to first cardiovascular event (myocardial infarction, stroke, coronary heart disease, congestive heart failure, or peripheral vascular disease). Secondary outcomes included osteoporosis, bone fracture, and a composite event (any of these). Cox models assessed the association between ADT as a time-dependent covariate and time-to-event across all patients by race.

RESULTS Among 790 916 veterans with PC (30 139 [4%] Hispanic; 145 267 non-Hispanic Black [18%]; 598 574 non-Hispanic White [76%]; and 16 936 other races [2%]), ADT was associated with increased risk of cardiovascular events (hazard ratio [HR], 1.16; 95% CI, 1.15-1.17; $P < .001$), bone fractures (HR, 1.59; 95% CI, 1.55-1.62; $P < .001$), osteoporosis (HR, 2.73; 95% CI, 2.65-2.80; $P < .001$), and composite event (HR, 1.29; 95% CI, 1.27-1.30; $P < .001$). When stratified by race, ADT remained significantly associated with all outcomes among all races. Among Black patients, compared with White patients, the excess risk due to ADT was slightly lower for cardiovascular events (HR, 1.09 [95% CI, 1.06-1.17] vs 1.18 [95% CI, 1.16-1.19]) and bone fracture (HR, 1.35 [95% CI, 1.28-1.42] vs 1.65 [95% CI, 1.61-1.70]) but higher for osteoporosis (HR, 3.16 [95% CI, 2.95-3.38] vs 2.64 [95% CI, 2.56-2.73]). HRs for Hispanic patients and those of other races were very similar to those for White patients.

CONCLUSIONS AND RELEVANCE In this cohort study of 790 916 veterans with PC, ADT was associated with increased cardiovascular and bone risks across all racial groups. Although excess risks

(continued)

Key Points

Question Do adverse events after receipt of androgen deprivation therapy differ by race?

Findings In this cohort study of 790 916 patients with prostate cancer, androgen deprivation therapy was associated with significantly increased risks of cardiovascular events, bone fractures, osteoporosis, and composite events across all racial groups. Although excess risks differed slightly between Black and White patients, overall differences were small.

Meaning Androgen deprivation therapy is associated with adverse cardiovascular and bone complications regardless of race, suggesting that current monitoring recommendations should not differ by race.

+ Supplemental content

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Abstract (continued)

differed slightly between Black and White patients, overall differences were small. These findings highlight the need for monitoring and mitigation strategies for patients receiving ADT.

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Introduction

Despite recent advances, many patients will still require systemic treatment for their prostate cancer (PC). Androgen deprivation therapy (ADT) remains the backbone of first-line systemic treatment for PC.¹ Although ADT can improve survival, it is also associated with substantial morbidity,² including increased risks of osteoporosis, diabetes, sexual dysfunction, and cardiovascular events, such as coronary heart disease, myocardial infarction, and sudden cardiac death.³⁻⁵

The extent to which these ADT-associated risks differ by race remains unclear because of limited research.⁶ Health disparities in PC are well documented, with Black patients experiencing higher rates of more aggressive disease and worse outcomes compared with White patients.^{7,8} Moreover, Black patients have a higher incidence of coronary artery disease,⁹ which could influence ADT-associated cardiovascular events. Within the Veterans Affairs (VA), however, studies suggest that Black patients have lower coronary artery disease incidence than White patients,^{10,11} likely reflecting the mitigating effect of equal access to health care. In contrast, for osteoporosis, Black individuals tend to have higher bone mineral density (BMD) and lower fracture risk than White individuals.¹² Given that many patients with advanced PC requiring ADT are from minoritized groups, it is important to determine whether race modifies these associations.

To examine this, we used data from the VA Health Care System (VAHCS) to perform a retrospective population-level analysis assessing whether race modifies the association between ADT and adverse events. On the basis of a prior modestly sized study suggesting that Black patients are more likely to experience cardiovascular events after ADT,⁶ we hypothesized that the magnitude of the association between ADT and adverse events would be greater among Black patients than those of other races, but smaller for osteoporosis, given the higher BMD in Black patients.

Methods

Design and Cohort

This nationwide, retrospective, cohort study used the VA Informatics and Computing Infrastructure, an analytic platform with access to all electronic health record data in the VAHCS. We used specific queries within VA Informatics and Computing Infrastructure to identify veterans who received a diagnosis of PC between January 2001 through December 2021 who were aged 18 years or older at the time of diagnosis and were considered active users of the VAHCS (defined as a minimum of 2 visits within 5 years of the study period). Demographic and clinical data were extracted from VA Informatics and Computing Infrastructure, including the type and dates of common PC-specific treatments. Patients were excluded if they did not have pertinent data on race and/or ethnicity. Patients were also excluded from analysis of a given end point if they experienced the event of interest or a censoring event (described below) on the same day as PC diagnosis, thus resulting in no follow-up time. The study was approved by the Durham VAHCS institutional review board, which permitted access to medical records for veterans with PC nationwide. Informed consent was waived because of the retrospective nature of the study and use of deidentified data. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines for cohort studies.

Study Variables

The primary variable of interest was receipt of ADT, and race and/or ethnicity was tested as an effect modifier. Race and ethnicity were both self-reported and combined into groups of Hispanic (regardless of race), non-Hispanic Black, non-Hispanic White, and other, which includes Asian, biracial, Native Hawaiian or Other Pacific Islander, and Native American or Alaska Native. ADT was defined as receipt of a luteinizing hormone-releasing hormone agonist (leuprolide, goserelin, or triptorelin) as a first-line systemic therapy and was captured as a time-dependent binary variable. The no-ADT group was defined as patients who did not receive ADT at any point during follow-up. Because of small numbers, patients who received nonsteroidal antiandrogens as a first-line therapy alone for more than 90 days without addition of a luteinizing hormone-releasing hormone agonist, undergoing bilateral orchiectomy as a first-line therapy alone, or receipt of a gonadotropin-releasing hormone (GnRH) antagonist as a first-line therapy alone for more than 90 days without addition of a luteinizing hormone-releasing hormone agonist were censored at receipt of that given therapy. GnRH antagonists (degarelix and relugolix) were introduced later in the study period and were used infrequently in our cohort, limiting their inclusion in the analysis. Patients were also censored at time of systemic second-line PC therapy (cabazitaxel, docetaxel, darolutamide, enzalutamide, abiraterone acetate, apalutamide, olaparib, rucaparib, pembrolizumab, radium-223, or sipuleucel-T) since these can impact the risk of events of interest. The primary outcome was time from PC diagnosis to a cardiovascular event, defined as the first myocardial infarction, stroke, coronary heart disease, heart failure, or peripheral vascular disease. Secondary outcomes included time from PC diagnosis to osteoporosis, bone fracture, and a composite outcome consisting of the first of any events above (cardiovascular, osteoporosis, or bone fracture). For all outcomes, patients without the event were censored at time of last known follow-up within the VA system, death, or receipt of second-line therapy described above (whichever came first).

Statistical Analysis

Data analysis was performed from January to March 2023. Baseline demographic and clinical characteristics of patients at PC diagnosis were summarized using standard descriptive statistics, stratified by race, ethnicity, and ADT use. Within each racial or ethnic group, differences in characteristics between ADT groups were assessed using independent samples *t* tests for continuous variables and χ^2 tests for categorical variables. Although *P* values are used to assess whether differences are due to chance, they do not quantify the size of differences and are heavily influenced by sample size. A large sample (as in our case) can lower the *P* value to highly significant even when differences are small and not clinically meaningful. Thus, we supplement *P* values (significance was defined as 2-sided *P* < .05) with standardized mean differences (SMDs), which are not influenced by sample size and compare across samples. We used an SAS (SAS Institute) macro developed by Yang et al¹³ that computes SMD as the difference in sample means or proportions divided by the SE and uses the Mahalanobis distance method to extend computation of SMD on categorical variables with *K* levels. Resulting effect sizes were expressed as absolute values. We considered absolute SMD greater than or equal to 0.1 as clinically relevant,¹⁴⁻¹⁶ as commonly used in PC studies.¹⁷

Univariable and multivariable Cox proportional hazards models were used to test the association between ADT and time to the given event. ADT was treated as a time-dependent covariate in all models. Other candidate variables included age at PC diagnosis (continuous), year of PC diagnosis (continuous), median annual household income of the zip code of residence (quartiles of less than or equal to \$39 578, \$39 580-\$48 879, \$48 883-\$62 578, \$62 679 or greater, and unknown), prostate-specific antigen (PSA) level at PC diagnosis (<4.0 ng/mL, 4.0-10.0 ng/mL, 10.1-19.9 ng/mL, $\geq$ 20.0 ng/mL, and unknown; to convert prostate-specific antigen to micrograms per liter, multiply by 1.0), Charlson Comorbidity Index (CCI) score at PC diagnosis (0, 1, 2, or $\geq$ 3), body mass index (calculated as weight in kilograms divided by height in meters squared) at PC diagnosis (<25, 25-29, $\geq$ 30, or unknown), receipt of radiation therapy (time-dependent covariate), undergoing radical prostatectomy (time-dependent covariate), and history of the given event at time of PC

diagnosis (yes or no). For example, history of cardiovascular events (or other events) was a variable in time to cardiovascular events (or other event) models. The final models included all candidate variables except for undergoing radical prostatectomy because of multicollinearity with PSA and it added little to no predictive value to the model above and beyond other covariates. Models were performed across all patients, and because results may vary by race and ethnicity, we also stratified the models by race and ethnicity to evaluate the association between ADT and time to events within each group separately. The interaction between ADT and race and ethnicity was tested for significance in multivariable models.

As patients who received ADT following primary localized therapy may have less advanced disease vs those who received ADT as first-line treatment and, thus, may respond to ADT differently, sensitivity analyses explored associations between ADT (vs no ADT) and risk of adverse events separately for patients who started ADT following localized therapy (radical prostatectomy or radiation) and when ADT was the first treatment received. All other modeling aspects were identical. Because history of cardiovascular and other events may impact likelihood of future events, sensitivity analyses explored limiting models to subgroups without history of the given event.

Results

Baseline Demographic and Clinical Characteristics

A total of 989 931 patients were identified, of whom 810 247 had known race and ethnicity. After excluding patients with no follow-up, 790 916 were left, of whom 30 139 (4%) were Hispanic, 145 267 (18%) were non-Hispanic Black, 598 574 (76%) were non-Hispanic White, and 16 936 (2%) were other races.

Because large sample sizes led to many statistically significant differences between ADT groups with regard to baseline and clinical characteristics (**Table 1** and **Table 2**), we describe only those that were clinically meaningful ($SMD \geq 0.1$) below. In general, age did not differ between patients who did or did not receive ADT, except among Hispanic patients, where those who received ADT were slightly older at diagnosis than those who did not receive ADT (median [IQR] age, 71.9 [65.3-77.7] vs 69.5 [63.0-76.5] years; SMD , 0.20). Patients with ADT received diagnoses of PC in slightly earlier years of the study compared with patients without ADT among non-Hispanic Black patients (median [IQR], 2011 [2006-2016] vs 2012 [2007-2017]; SMD , 0.13) and Hispanic patients (median [IQR], 2009 [2005-2015] vs 2011 [2006-2016]; SMD , 0.21). Median zip code income levels were significantly lower for patients with ADT compared with patients without ADT across all racial and ethnic groups ($SMDs$, 0.13 to 0.28). Patients with ADT were more likely to have 3 or more comorbidities compared with patients without ADT across all racial and ethnic groups ($SMDs$, 0.12 to 0.22). In general, history of cardiovascular and other events did not differ between ADT groups across all racial and ethnic groups. The majority of patients were missing clinical stage and Gleason score, but clinically meaningful differences in other PC characteristics were observed between ADT groups. Patients who received ADT had significantly higher PSA levels at PC diagnosis than patients who did not receive ADT across all racial and ethnic groups ($SMDs$, 0.38 to 0.44). Patients with ADT were much more likely to also have received radiation during the study compared with patients without ADT across all racial and ethnic groups ($SMDs$, 0.74 to 0.87). Radical prostatectomy rates were lower among patients who received ADT compared with patients who did not receive ADT among non-Hispanic Black and Hispanic patients ($SMDs$, 0.13 to 0.10) (Table 2).

Primary Outcome: Cardiovascular Events

During a median (IQR) follow-up of 79.9 (36.1-137.2) months, there were 289 405 (37%) cardiovascular events. Across all patients and after adjusting for race and ethnicity, ADT was associated with significantly increased cardiovascular events risk in univariable (hazard ratio [HR], 1.36; 95% CI, 1.35-1.37) and multivariable (HR, 1.16; 95% CI, 1.15-1.17; $P < .001$) analyses. Although interactions between race and ethnicity and ADT were statistically significant, the magnitude of

Table 1. Patient Demographic and Clinical Characteristics by ADT Status

	Patients, No. (%)			
Characteristic	All patients (N = 790 916)	No ADT (n = 640 000)	ADT (n = 150 916)	SMD ^a
Race and ethnicity				
Hispanic	30 139 (4)	22 433 (4)	7706 (5)	0.25 ^c
Non-Hispanic Black	145 267 (18)	106 401 (17)	38 866 (26)	
Non-Hispanic White	598 574 (76)	497 753 (78)	100 821 (67)	
Other ^b	16 936 (2)	13 413 (2)	3523 (2)	
Age at PC diagnosis, median (IQR), y	71.0 (64.5-77.8)	71.2 (64.6-78.0)	70.2 (63.9-76.9)	0.08 ^c
Year of PC diagnosis, median (IQR)	2010 (2005-2015)	2010 (2005-2016)	2010 (2005-2015)	0.02 ^c
Annual household income				
Missing, No.	125 520	97640	27 880	0.20 ^c
Median (IQR), \$	48 663 (39 439-62 307)	49 290 (39 949-63 032)	46 267 (37 081-58 542)	
Body mass index at PC diagnosis ^d				
Missing, No.	132 342	105 248	27 094	0.08 ^c
<25	150 402 (23)	120 397 (23)	30 005 (24)	
25-29	276 887 (42)	228 186 (43)	48 701 (39)	
≥30	231 285 (35)	186 169 (35)	45 116 (36)	
Clinical stage				
Missing, No.	724 162	600 484	123 678	0.41 ^c
T1	43 522 (65)	28 259 (72)	15 263 (56)	
T2	19 827 (30)	10 347 (26)	9480 (35)	
T3	2547 (4)	776 (2)	1771 (7)	
T4	858 (1)	134 (0)	724 (3)	
Prostate-specific antigen level at PC diagnosis, ng/mL				
<4.0	186 977 (24)	169 590 (26)	17 387 (12)	0.87 ^c
4.0-10.0	177 651 (22)	131 569 (21)	46 082 (31)	
10.1-19.9	50 324 (6)	27 795 (4)	22 529 (15)	
≥20.0	38 213 (5)	12 433 (2)	25 780 (17)	
Unknown	337 751 (43)	298 613 (47)	39 138 (26)	
Grade at PC diagnosis (Gleason score)				
Missing, No.	699 488	584 957	114 531	1.05 ^c
Grade group 1 (2-6)	32 575 (36)	26 740 (49)	5835 (16)	
Grade group 2 (3 + 4)	26 752 (29)	17 700 (32)	9052 (25)	
Grade group 3 (4 + 3)	11 834 (13)	5818 (11)	6016 (17)	
Grade group 4 (8)	10 380 (11)	3056 (6)	7324 (20)	
Grade group 5 (9-10)	9887 (11)	1729 (3)	8158 (22)	
Charlson Comorbidity Index score at PC diagnosis				
0	389 978 (49)	328 757 (51)	61 221 (41)	0.20 ^c
1	133 298 (17)	104 403 (16)	28 895 (19)	
2	80 755 (10)	63 020 (10)	17 735 (12)	
≥3	186 885 (24)	143 820 (22)	43 065 (29)	
Medical history before PC				
Cardiovascular disease ^e				
No	512 388 (65)	415 840 (65)	96 548 (64)	0.02 ^c
Yes	278 528 (35)	224 160 (35)	54 368 (36)	
Diabetes				
No	712 652 (90)	579 937 (91)	132 715 (88)	0.09 ^c
Yes	78 264 (10)	60 063 (9)	18 201 (12)	
Hypertension				
No	223 470 (28)	184 386 (29)	39 084 (26)	0.07 ^c
Yes	567 446 (72)	455 614 (71)	111 832 (74)	

(continued)

Table 1. Patient Demographic and Clinical Characteristics by ADT Status (continued)

Characteristic	Patients, No. (%)			SMD ^a
	All patients (N = 790 916)	No ADT (n = 640 000)	ADT (n = 150 916)	
Osteoporosis				
No	771 736 (98)	624 614 (98)	147 122 (97)	0.01 ^f
Yes	19 180 (2)	15 386 (2)	3794 (3)	
Bone fracture				
No	756 673 (96)	614 484 (96)	142 189 (94)	0.08 ^c
Yes	34 243 (4)	25 516 (4)	8727 (6)	
Radiation treatment				
No	673 848 (85)	584 767 (91)	89 081 (59)	0.81 ^c
Yes	117 068 (15)	55 233 (9)	61 835 (41)	
Radical prostatectomy				
No	738 626 (93)	597 837 (93)	140 789 (93)	0.00
Yes	52 290 (7)	42 163 (7)	10 127 (7)	

Abbreviations: ADT, androgen deprivation therapy; PC, prostate cancer; SMD, standardized mean difference.

SI conversion factor: To convert prostate-specific antigen to micrograms per liter, multiply by 1.0.

^a SMDs are calculated as the difference in means or proportions divided by the SE. All SMDs are expressed as absolute values.

^b Other refers to Asian, biracial, Native Hawaiian or Other Pacific Islander, or Native American or Alaska Native.

^c $P < .001$.

^d Body mass index is calculated as weight in kilograms divided by height in meters squared.

^e Cardiovascular disease refers to myocardial infarction, stroke, coronary heart disease, congestive heart failure, and/or peripheral heart disease.

^f $P < .05$.

differences was small (univariable HRs ranging from 1.30 to 1.39, and multivariable HRs ranging from 1.09 to 1.18), although excess risks were slightly lower in Black (HR, 1.09; 95% CI, 1.06-1.17) vs White (HR, 1.18; 95% CI, 1.16-1.19) patients (**Table 3**).

Secondary Outcomes: Osteoporosis, Bone Fractures, and Composite Event

During the same follow-up time, there were 62 944 (8%) bone fracture events, 34 454 (4%) osteoporosis events, and 326 904 (41%) total composite events. Similar to the primary outcome, across all patients, ADT was associated with significantly increased risk for all secondary outcomes examined in both univariable and multivariable analysis. ADT was most associated with osteoporosis (multivariable HR, 2.73; 95% CI, 2.65-2.80; $P < .001$), followed by bone fracture (multivariable HR, 1.59; 95% CI, 1.55-1.62; $P < .001$) and the composite outcome (multivariable HR, 1.29; 95% CI, 1.27-1.30; $P < .001$). When stratified by race and ethnicity, the positive association between ADT and each outcome remained similar across all races and ethnicities, although the magnitude of the association varied slightly in some instances. For example, the association between ADT and increased osteoporosis risk was greatest among non-Hispanic Black patients (multivariable HR, 3.16 [95% CI, 2.95-3.38] vs 2.64 [95% CI, 2.56-2.73] among White patients) compared with other groups (HRs ≤ 2.79), although the association between ADT and increased bone fractures risk was lowest among non-Hispanic Black patients (multivariable HR, 1.35 [95% CI, 1.28-1.42] vs 1.65 [95% CI, 1.61-1.70] among White patients). The association between ADT and occurrence of any event was very similar across all races and ethnicities (HRs ranging from 1.27 to 1.36) except non-Hispanic Black patients, for whom associations were slightly smaller (multivariable HR, 1.19; 95% CI, 1.16-1.22). Similar to the primary outcome, interactions between race and ethnicity and ADT were statistically significant for all secondary outcomes, although again the magnitude of differences was small (Table 3).

Sensitivity Analyses

When patients were stratified by primary ADT vs ADT after local therapy, across both subsets, associations between ADT and risk of cardiovascular, bone fracture, osteoporosis, and composite

Table 2. Patient Demographic and Clinical Characteristics by Race and Receipt of ADT (N = 790 916)

Characteristic	Hispanic			Non-Hispanic Black			Non-Hispanic White			Other ^a		
	Patients, No. (%)			Patients, No. (%)			Patients, No. (%)			Patients, No. (%)		
	No ADT (n = 22 433)	ADT (n = 7706 [26%])		No ADT (n = 106 401)	ADT (n = 38 866 [27%])		No ADT (n = 497 753)	ADT (n = 100 821 [17%])		No ADT (n = 13 413)	ADT (n = 3523 [21%])	
Age at PC diagnosis, median (IQR), y	69.5 (63.0-76.5)	71.9 (65.3-77.7)	SMD ^b 0.20 ^c	65.7 (59.8-72.5)	66.5 (60.8-73.2)	0.09 ^c	72.4 (66.0-78.8)	71.4 (65.2-77.9)	0.07 ^c	70.0 (63.6-77.1)	69.9 (63.7-76.6)	0.01
Year of PC diagnosis, median (IQR)	2011 (2006-2016)	2009 (2005-2015)	0.21 ^c	2012 (2007-2017)	2011 (2006-2016)	0.13 ^c	2010 (2005-2015)	2010 (2005-2015)	0.01 ^d	2011 (2006-2016)	2011 (2006-2016)	0.03
Annual household income												
Missing, No.	3872	1693		15 753	6831		75 834	18 672		2181	684	
Median (IQR), \$	40 614 (24 961-55 152)	32 548 (20 450-48 775)	0.28 ^c	42 581 (32 884-56 273)	39 917 (31 418-51 971)	0.17 ^c	50 831 (41 679-64 655)	48 800 (40 516-61 250)	0.13 ^c	52 333 (40 921-68 795)	48 891 (38 284-62 648)	0.19 ^c
Body mass index at PC diagnosis ^e												
Missing, No.	3968	1327		20 190	7176		78 497	17 908		2593	683	
<25	4462 (24)	1766 (28)		21 180 (25)	9016 (28)		91 976 (22)	18 493 (22)		2779 (26)	730 (26)	
25-29	8118 (44)	2667 (42)	0.09 ^c	32 277 (37)	11 266 (36)	0.07 ^c	183 178 (44)	33 637 (41)	0.07 ^c	4613 (43)	1131 (40)	0.07 ^d
≥30	5885 (32)	1946 (31)		32 754 (38)	11 408 (36)		144 102 (34)	30 783 (37)		3428 (32)	979 (34)	
Clinical stage												
Missing, No.	20150	5554		95 081	30 413		472 688	84 700		12 565	3011	
T1	1176 (52)	787 (37)		8765 (77)	5407 (64)		17 707 (71)	8785 (54)		611 (72)	284 (55)	
T2	1028 (45)	1175 (55)	0.38 ^c	2328 (21)	2356 (28)	0.37 ^c	6771 (27)	5767 (36)	0.44 ^c	220 (26)	182 (36)	0.43 ^c
T3	65 (3)	117 (5)		195 (2)	461 (5)		501 (2)	1164 (7)		15 (2)	29 (6)	
T4	14 (1)	73 (3)		32 (0)	229 (3)		86 (0)	405 (3)		2 (0)	17 (3)	
Prostate-specific antigen at PC diagnosis, ng/mL												
<4.0	6099 (27)	871 (11)		26 100 (25)	3191 (8)		134 218 (27)	12 960 (13)		3173 (24)	365 (10)	
4.0-10.0	5639 (25)	2535 (33)		32 045 (30)	12 499 (32)		90 975 (18)	30 017 (30)		2910 (22)	1031 (29)	
10.1-19.9	1223 (5)	1232 (16)	0.85 ^c	7175 (7)	6635 (17)	0.86 ^c	18 736 (4)	14 134 (14)	0.83 ^c	661 (5)	528 (15)	0.80 ^c
≥20.0	522 (2)	1279 (17)		3420 (3)	8527 (22)		8212 (2)	15 381 (15)		279 (2)	593 (17)	
Unknown	8950 (40)	1789 (23)		37 661 (35)	8014 (21)		245 612 (49)	28 329 (28)		6390 (48)	1006 (29)	
Grade at PC diagnosis (Gleason score)												
Missing, No.	20 096	5976		90 367	27 478		462 276	78 353		12 218	2724	
Grade group 1 (2-6)	1227 (53)	348 (20)		7351 (46)	1936 (17)		17 622 (50)	3429 (15)		540 (45)	122 (15)	
Grade group 2 (3 + 4)	671 (29)	457 (26)		5556 (35)	3188 (28)		11 061 (31)	5194 (23)		412 (34)	213 (27)	
Grade group 3 (4 + 3)	244 (10)	277 (16)	0.95 ^c	1789 (11)	1981 (17)	0.92 ^c	3651 (10)	3637 (16)	1.11 ^c	134 (11)	121 (15)	0.96 ^c
Grade group 4 (8)	130 (6)	340 (20)		872 (5)	2174 (19)		1987 (6)	4667 (21)		67 (6)	143 (18)	
Grade group 5 (9-10)	65 (3)	308 (18)		466 (3)	2109 (19)		1156 (3)	5541 (25)		42 (4)	200 (25)	
Charlson Comorbidity Index score at PC diagnosis												
0	9319 (42)	2613 (34)		44 141 (4,1)	13 498 (35)		268 978 (54)	43 742 (43)		6319 (47)	1368 (39)	
1	4102 (18)	1585 (21)	0.17 ^c	19140 (18)	7677 (20)	0.12 ^c	78 811 (16)	18 955 (19)	0.22 ^c	2350 (18)	678 (19)	0.17 ^c
2	2380 (11)	972 (13)		11 456 (11)	4680 (12)		47 847 (10)	11 650 (12)		1337 (10)	433 (12)	
≥3	6632 (30)	2536 (33)		31 664 (30)	13 011 (33)		102 117 (21)	26 474 (26)		3407 (25)	1044 (30)	

(continued)

Table 2. Patient Demographic and Clinical Characteristics by Race and Receipt of ADT (N = 790 916) (continued)

Characteristic	Hispanic		Non-Hispanic Black		Non-Hispanic White		Other ^a	
	Patients, No. (%)		Patients, No. (%)		Patients, No. (%)		Patients, No. (%)	
	No ADT (n = 22 433)	ADT (n = 7706 [26%])	No ADT (n = 106 401)	ADT (n = 38 866 [27%])	No ADT (n = 497 753)	ADT (n = 100 821 [17%])	No ADT (n = 13 413)	ADT (n = 3523 [21%])
Medical history before PC								
Cardiovascular disease ^f								
No	15 204 (68)	5073 (66)	74 681 (70)	26 317 (68)	316 971 (64)	62 859 (62)	8984 (67)	2299 (65)
Yes	7229 (32)	2633 (34)	31 720 (30)	12 549 (32)	180 782 (36)	37 962 (38)	4429 (33)	1224 (35)
Diabetes								
No	19 137 (85)	6448 (84)	91 583 (86)	32 954 (85)	457 390 (92)	90 294 (90)	11 827 (88)	3019 (86)
Yes	3296 (15)	1258 (16)	14 818 (14)	5912 (15)	40 363 (8)	10 527 (10)	1586 (12)	504 (14)
Hypertension								
No	5902 (26)	1711 (22)	21 467 (20)	7455 (19)	153 479 (31)	29 047 (29)	3538 (26)	871 (25)
Yes	16 531 (74)	5995 (78)	84 934 (80)	31 411 (81)	344 274 (69)	71 774 (71)	9875 (74)	2652 (75)
Osteoporosis								
No	21 780 (97)	7438 (97)	104 924 (99)	38 263 (98)	484 824 (97)	98 003 (97)	13 086 (98)	3418 (97)
Yes	653 (3)	268 (3)	1477 (1)	603 (2)	12 929 (3)	2818 (3)	327 (2)	105 (3)
Bone fracture								
No	21 230 (95)	7259 (94)	101 035 (95)	36 396 (94)	479 362 (96)	95 203 (94)	12 857 (96)	3331 (95)
Yes	1203 (5)	447 (6)	5366 (5)	2470 (6)	18 391 (4)	5618 (6)	556 (4)	192 (5)
Radiation								
No	20 071 (89)	4575 (59)	89 661 (84)	17 915 (46)	462 904 (93)	64 461 (64)	12 131 (90)	2130 (60)
Yes	2362 (11)	3131 (41)	16 740 (16)	20 951 (54)	34 849 (7)	36 360 (36)	1282 (10)	1393 (40)
Radical prostatectomy								
No	20 030 (89)	7100 (92)	94 035 (88)	35 809 (92)	471 251 (95)	94 585 (94)	12 521 (93)	3295 (94)
Yes	2403 (11)	606 (8)	12 366 (12)	3057 (8)	26 502 (5)	6236 (6)	892 (7)	228 (6)

Abbreviations: ADT, androgen deprivation therapy; PC, prostate cancer; SMD, standardized mean difference.

SI conversion factor: To convert prostate-specific antigen to micrograms per liter, multiply by 1.0.

^a Other race includes Asian, biracial, Native Hawaiian or Other Pacific Islander, and Native American and Alaska Native races.

^b SMD was calculated as the difference in means or proportions divided by the SE. All SMDs were expressed as absolute values.

^c $P < .001$.

^d $P < .05$.

^e Body mass index is calculated as weight in kilograms divided by height in meters squared.

^f Cardiovascular disease refers to myocardial infarction, stroke, coronary heart disease, congestive heart failure, and/or peripheral heart disease.

events were similar (eTable 1 in [Supplement 1](#)). Although associations were slightly greater in the initial ADT group (except osteoporosis, which was greater in the ADT after local therapy), differences were minimal. Importantly, patterns remained similar across races.

Among patients without a history of the given event, associations between ADT and risk of all outcomes were similar to (and in many cases greater than) those in our primary analysis (eTable 2 in [Supplement 1](#)). Importantly, patterns were similar across races.

Discussion

Many prior studies linked ADT with adverse events such as cardiac events and osteoporosis.^{4,18,19} To date, there are limited studies addressing whether these associations differ by race.⁶ To address this, we performed a population-level cohort study using retrospective data from the nationwide VAHCS to compare racial differences in adverse events associated with ADT. Using a cohort of over 790 000 patients, we found significantly increased risks associated with ADT, including cardiovascular events, osteoporosis, and bone fractures, across all racial and ethnic groups. However, the extent of increased risk varied slightly among different populations, indicating that some groups may be more vulnerable than others to certain adverse events. For instance, the excess risk due to ADT was lower in Black patients for cardiovascular events (HR, 1.09 vs 1.18) and bone fractures (HR, 1.35 vs 1.65) compared with White patients but was higher for osteoporosis (HR, 3.16 vs 2.64). However, the clinical relevance of these slight variations in the extent of increased risk is unclear. These findings

Table 3. Univariable and Multivariable Associations Between ADT as a Time-Varying Variable and Time to Events Overall and by Race

Event and analysis	HR (95% CI) ^a				
	All patients (N = 790 916)	Hispanic (n = 30 139)	Non-Hispanic Black (n = 145 267)	Non-Hispanic White (n = 598 574)	Other (n = 16 936) ^b
Cardiovascular ^c					
Univariable	1.36 (1.35-1.37)	1.38 (1.32-1.44)	1.30 (1.27-1.33)	1.39 (1.37-1.40)	1.39 (1.30-1.48)
Multivariable ^d	1.16 (1.15-1.17)	1.18 (1.12-1.24)	1.09 (1.06-1.17)	1.18 (1.16-1.19)	1.14 (1.06-1.23)
Bone fracture					
Univariable	2.11 (2.07-2.14)	1.84 (1.70-1.98)	1.65 (1.59-1.72)	2.31 (2.26-2.36)	2.24 (1.99-2.52)
Multivariable ^e	1.59 (1.55-1.62)	1.48 (1.35-1.61)	1.35 (1.28-1.42)	1.65 (1.61-1.70)	1.75 (1.52-2.01)
Osteoporosis					
Univariable	2.76 (2.69-2.82)	3.13 (2.86-3.43)	3.40 (3.21-3.61)	2.74 (2.67-2.81)	2.42 (2.07-2.84)
Multivariable ^f	2.73 (2.65-2.80)	2.79 (2.52-3.10)	3.16 (2.95-3.38)	2.64 (2.56-2.73)	2.29 (1.91-2.76)
Composite ^g					
Univariable	1.52 (1.51-1.54)	1.59 (1.53-1.67)	1.42 (1.40-1.45)	1.57 (1.55-1.58)	1.55 (1.46-1.65)
Multivariable ^h	1.29 (1.27-1.30)	1.36 (1.30-1.43)	1.19 (1.16-1.22)	1.31 (1.29-1.33)	1.27 (1.19-1.36)

Abbreviations: ADT, androgen deprivation therapy; BMI, body mass index; CCI, Charlson Comorbidity Index; HR, hazard ratio; PC, prostate cancer; PSA, prostate-specific antigen level.

^a All comparisons are for ADT vs no ADT. For all associations, $P < .001$.

^b Other includes Asian, biracial, Native Hawaiian or Other Pacific Islander, and Native American and Alaska Native races.

^c Cardiovascular events refer to myocardial infarction, stroke, coronary heart disease, congestive heart failure, or peripheral heart disease.

^d Adjusted for age at PC, year of PC, race, PSA, CCI score, BMI, median household income, ADT (time-dependent), radiation (time-dependent), and history of cardiovascular events.

^e Adjusted for age at PC, year of PC, PSA, CCI score, BMI, median household income, ADT (time-dependent), radiation (time-dependent), and history of bone fracture.

^f Adjusted for age at PC, year of PC, PSA, CCI score, BMI, median household income, ADT (time-dependent), radiation (time-dependent), and history of osteoporosis.

^g Composite event of first of any cardiovascular, bone fracture, or osteoporosis events.

^h Adjusted for age at PC, year of PC, PSA, CCI score, BMI, median household income, ADT (time-dependent), radiation (time-dependent), and history of any event above.

underscore the importance of monitoring and managing these risks in patients undergoing ADT, regardless of race.

Cardiovascular disease is the leading cause of death among men with PC, and multiple studies have linked ADT to increased cardiovascular morbidity and mortality.^{4,20,21} The Food and Drug Administration has also warned about its cardiotoxic effects.²² The rates at which different races are affected by PC and cardiovascular disease vary. It is of utmost importance, then, to investigate whether ADT magnifies baseline racial differences. Using a large veteran cohort, ADT was significantly associated with cardiovascular events across all races. Notably, the risk was slightly lower among Black compared with White patients, despite previous studies suggesting higher cardiovascular risk in Black patients due to underlying disparities in cardiovascular health and access to preventive care.⁶ Although a prior study²¹ reported that Hispanic patients had a lower cardiovascular risk than White patients, our results showed no significant difference in excess risk due to ADT between these groups. Importantly, the clinical relevance of these slight variations across racial groups is likely small, reinforcing that ADT is substantially associated with cardiovascular risk for all patients. The older age of our cohort reflects treatment in the general population, where younger men more often receive definitive local therapies and ADT is more common in older men. Although all *P* values for interaction reached statistical significance, this reflects the large sample size, because absolute differences between groups were small and likely clinically irrelevant. These findings highlight the importance of vigilant cardiac monitoring and preventive strategies—such as blood pressure control, lipid management, glucose monitoring, and early identification of ischemic symptoms—for all men receiving ADT to reduce cardiovascular morbidity and mortality.

In addition to cardiovascular risks, ADT greatly affects bone health by increasing bone turnover and resorption, leading to higher rates of fractures.²³ Interestingly, our study found that the extent of increased fracture risk associated with ADT was lower among Black patients when compared with other racial or ethnic groups. This finding may be explained by prior research showing that Black patients generally have higher baseline BMD at all skeletal sites and experience fewer prevalent fractures compared with White patients and, thus, may be slightly more resistant to the adverse effects of ADT.²⁴ Intriguingly, despite being perhaps more protected from the fracture risk of ADT, Black patients had higher rates of ADT-associated osteoporosis than White patients. Intuitively, the 2 should be linked, because osteoporosis is a major risk factor for fracture.²⁵ However, osteoporosis requires testing to confirm the diagnosis, and, thus, many patients may have undiagnosed osteoporosis. Prior studies of patients with PC undergoing routine BMD testing show osteoporosis rates of 35.4% to 41%.^{26,27} In contrast, our claims-based study found low prediagnosis rates, depending on race, indicating substantial underdiagnosis. Thus, the clinical implications of slight racial differences in osteoporosis remain unclear. Still, osteoporosis remains a critical consequence of ADT, as patients experience rapid BMD decline, up to 17.3% higher than non-ADT groups within the first 6 to 12 months of therapy.²⁸ Although some studies have examined the association between ADT and osteoporosis across different racial groups,²⁹ gaps remain in understanding racial differences in osteoporosis risk. These data support that monitoring BMD, assessing fracture risk, and screening for and treating osteoporosis are crucial in patients receiving ADT. Our findings reinforce what is already well established; ADT is associated with increased cardiovascular and bone-related risks. Although we observed modest racial differences, their clinical relevance appears limited. Clinically, the message is clear—vigilance for ADT-associated complications is warranted for all men, regardless of race.

Strengths and Limitations

One of the notable strengths of this study is its large, racially diverse sample size, which enhances statistical power and generalizability of the findings. This study includes a large number of non-Hispanic Black individuals, who are often underrepresented in PC research, further strengthening the generalizability and applicability of our results to this population. However, there are also limitations to consider. First, we lacked data on certain risk factors, including family history,

diet, and smoking status, nor did we adjust for medications that may alter risk of the events being examined. Additionally, PC stage and grade data were unavailable for many patients, which may have introduced confounding given ADT's greater use in high-risk disease. Second, whether these findings apply to nonveterans is unclear. The VA's more uniform access to care may partly explain the minimal racial differences observed. Third, censoring patients who received second-line therapies may have underestimated ADT effects, as these treatments could impact outcomes. We also did not include GnRH antagonists such as degarelix or relugolix in our primary analysis because of limited uptake during the study period, which may leave unanswered whether associations differ by ADT class. Also, race is a social construct rather than a biological category. Fourth, despite adjusting for covariates, unmeasured confounders may still affect the observed associations.

Conclusions

To our knowledge, this is the first population-level study assessing the association between ADT use and risks of cardiovascular, bone complications, osteoporosis, and composite events. In this study, we found that although the excess risks of ADT differed slightly between Black and White patients, overall differences were small. These findings demonstrate that ADT is associated with adverse cardiovascular and bone complications regardless of race.

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Author Contributions: Ms Janes had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Friedrich, Janes, Parrish, Josan, Tatonetti, Freedland.

Acquisition, analysis, or interpretation of data: Friedrich, Janes, Parrish, De Hoedt, Josan, Freedland.

Drafting of the manuscript: Friedrich, Janes, Tatonetti, Freedland.

Critical review of the manuscript for important intellectual content: Friedrich, Janes, Parrish, De Hoedt, Josan, Freedland.

Statistical analysis: Janes.

Obtained funding: Tatonetti.

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Supervision: Parrish, De Hoedt, Josan, Tatonetti, Freedland.

Conflict of Interest Disclosures: Dr Friedrich reported comarketing the studied drug with Pfizer, Inc, and having financial interest in the results. Ms Janes reported having a financial interest in the results with Pfizer, Inc, outside the submitted work. Mr Parrish reported receiving payments to his institution from Myovant Sciences, Astellas Pharmaceuticals, Cedars-Sinai, Janssen, Genomic Health, Exact Sciences, Delfi Diagnostics, Guardant Health, Photocure, Prostatype, Novartis, VIR Biotechnology, AstraZeneca, Merck, University of California, Los Angeles, Coloplas, Reinvestment Partners, LungLife AI, Pfizer, Myriad Genetics, Blue Earth Diagnostics, and Ferring outside

the submitted work. Ms De Hoedt reported receiving grants from Merck, Janssen, Astellas, and AstraZeneca outside the submitted work. Dr Freedland reported receiving personal fees from Astellas, Bayer, Pfizer, Sanofi, Merck, AstraZeneca, Eli Lilly, Novartis, Janssen, and Tolmar outside the submitted work. No other disclosures were reported.

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Data Sharing Statement: See [Supplement 2](#).

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SUPPLEMENT 1.

eTable 1. Univariable and multivariable associations between ADT as first-line (1L) therapy for PC as well as ADT as second-line (2L) therapy for PC (following primary localized therapy with RP or radiation) and time to events overall and by race

eTable 2. Univariable and multivariable associations between ADT (vs. no ADT) as a time-varying predictor and time to events among patients who did not have prior history of the given event at time of PC diagnosis

SUPPLEMENT 2.

Data Sharing Statement